

THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFARABAD

Department of Software Engineering

LAB 02

Submitted to: *Engr. Dr. Asma Javaid*

Session: *2024-28*

Roll No: *2024-SE-34*

Submitted by*: Jawahir Ali*

Date: *24-10-2026*

1. Activity Diagram Visualizing the Process of The Foodpanda Application (Online Food Order App) Using Swimlanes.

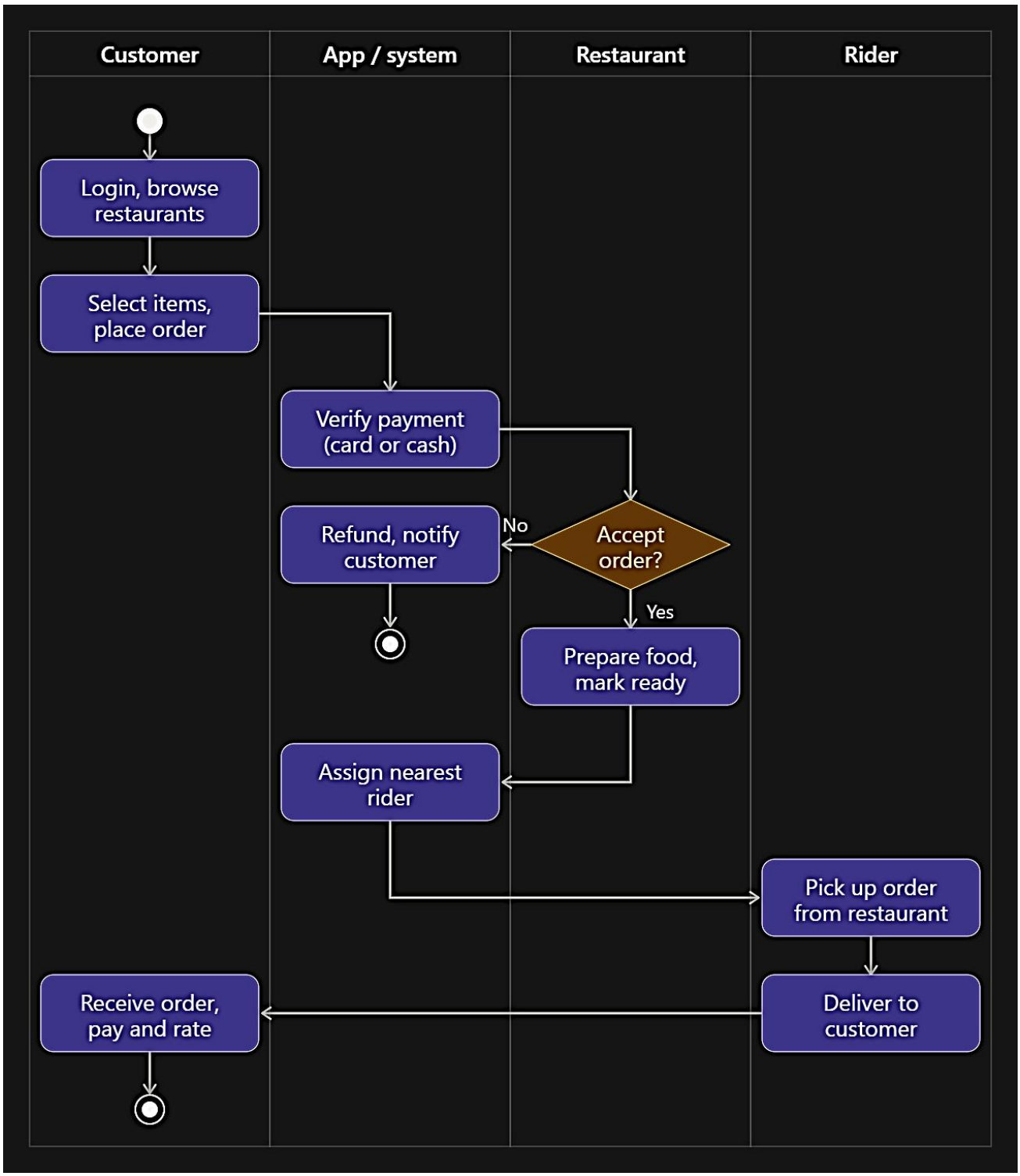


Figure 1

Description

This task models the complete online food ordering process of foodPanda as a UML activity diagram. Four swimlanes (Customer, App/System, Restaurant, and Rider) show which actor performs each activity. The flow starts when the customer browses restaurants and places an order. The system then verifies the payment, and the restaurant accepts or rejects the order. If the restaurant rejects it, the system refunds the customer and notifies them. If it is accepted, the food is prepared, a rider is assigned, and the rider picks up and delivers the order. The process ends when the customer receives the food and rates the service.

2. State Diagram for A Toaster.

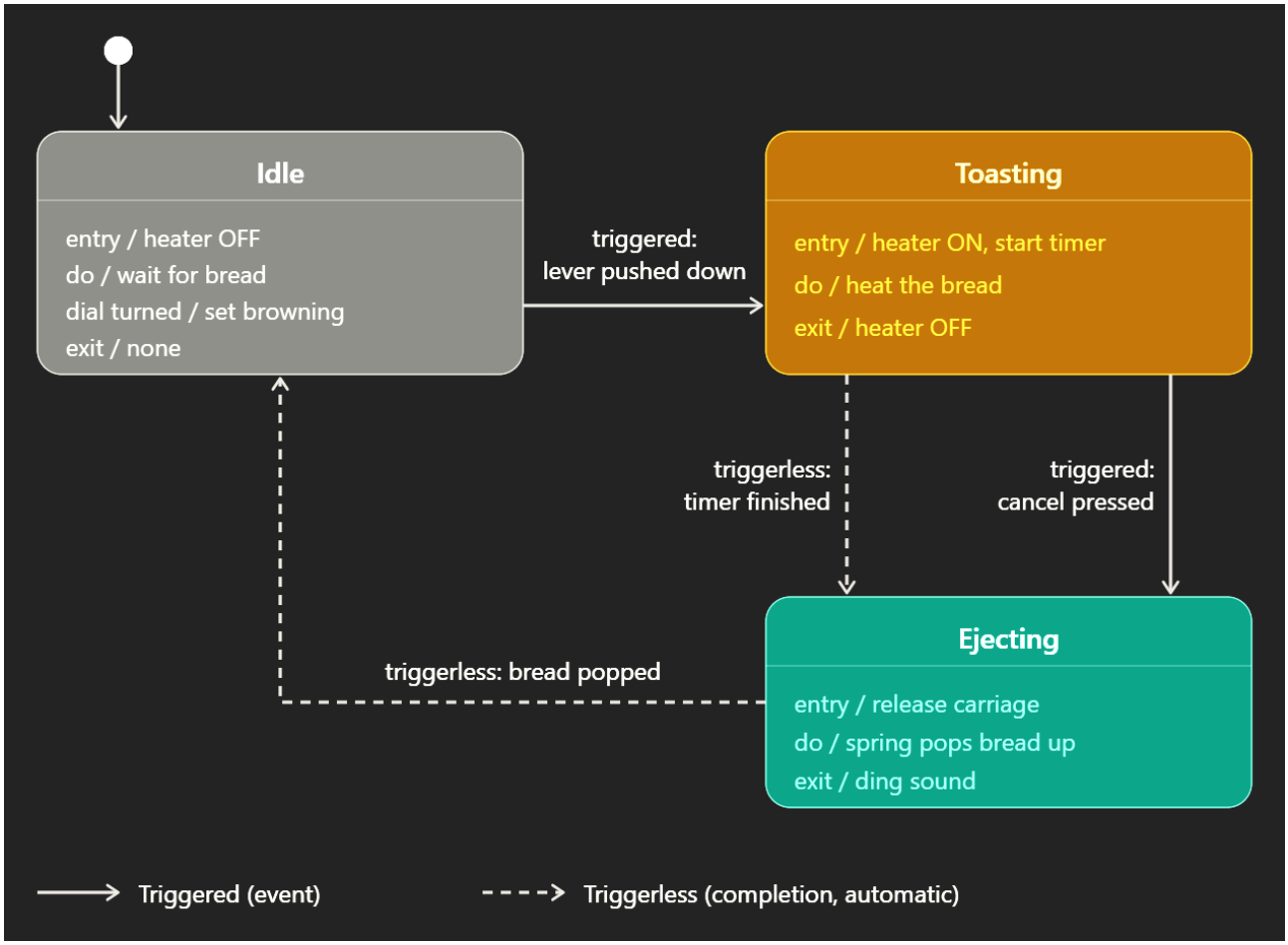


Figure 2

Description

This task models the behavior of a toaster as a UML state diagram with three states: Idle, Toasting, and Ejecting. The activities inside each state are listed as entry, do, and exit actions (for example, entry: heater ON, do: heat the bread, exit: heater OFF). The task also separates the two kinds of transitions:

- Triggered events happen because of an external event, such as pushing the lever down, pressing cancel, or turning the browning dial.
- Triggerless (completion) transitions happen automatically when a state's activity finishes, such as the timer ending or the bread popping up.

3. Draw A State Machine Diagram of an Elevator/Lift.

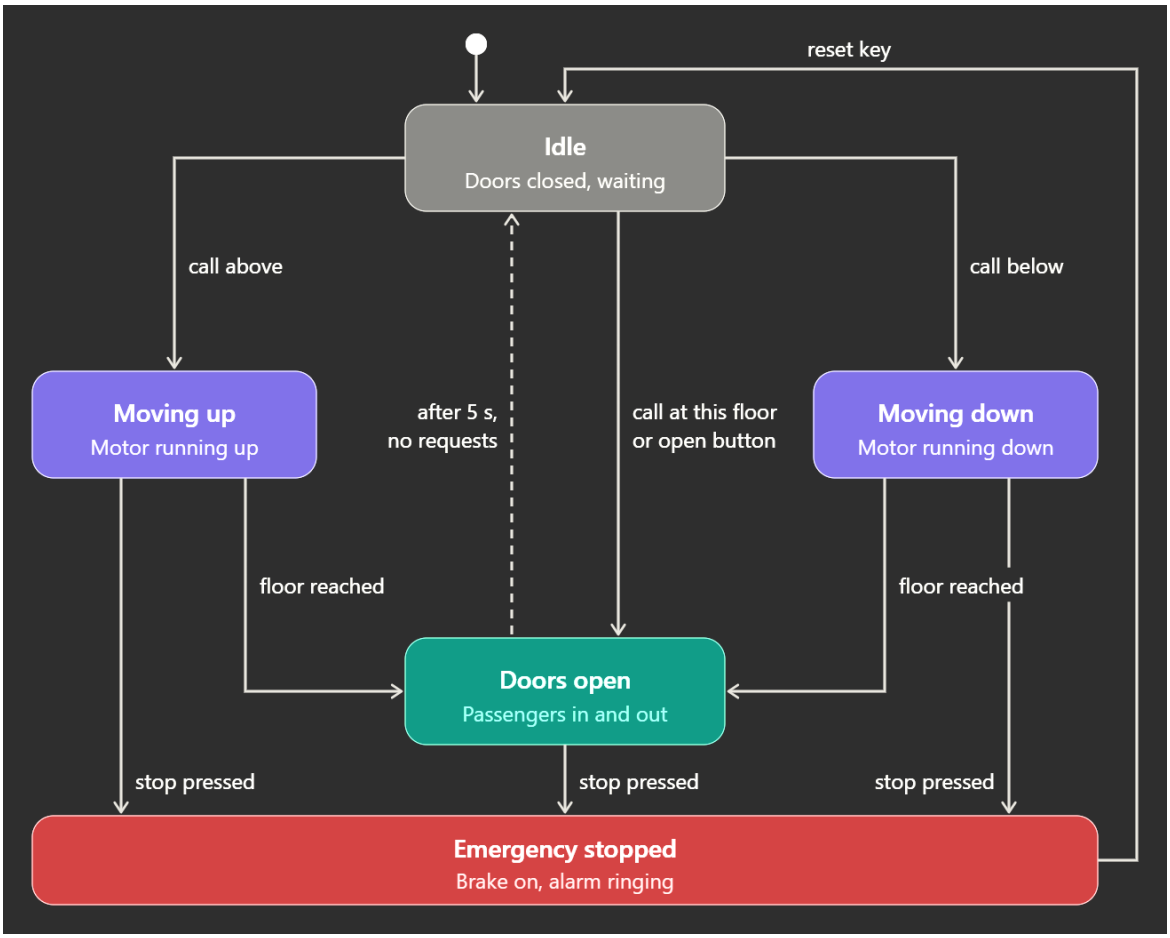


Figure 3

Description

This task models an elevator's behavior as a UML state machine with five states: ***Idle, Moving Up, Moving Down, Doors Open, and Emergency Stopped***. Transitions are caused by floor calls, the floor sensor reaching the target floor, the door-open timer expiring, the emergency stop button, and the reset key. The diagram shows how the lift moves between waiting, travelling, opening its doors, and stopping safely in an emergency**.**